Joseph C. Cuthbert

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EDUCATION

Georgia Institute of Technology — **Atlanta, Georgia**
Pursuing a Master's of Science Degree in Computational Science and Engineering — August 2025 — December 2026
Berry College — Class of 2025 — **Mount Berry, Georgia**
B.S. Computer Science and Mathematics — August 2021 — May 2025
Cumulative GPA: 3.76/4.00
Participation: NCAA Men's Lacrosse Team, National College Athlete Honor Society

EXPERIENCE

Sensors Directorate Intern — May 2026 — August 2026
University of Dayton, NASIC — Dayton, Ohio

- Researched the robustness of CNN classification models by exploring the effect of extraneous noise signals in SAR images
- Implemented data augmentation techniques in PyTorch to reduce model vulnerability to noise and improve generalization

Machine Learning Research Assistant — June 2023 — January 2025
Berry College Department of Computer Science — Mount Berry, Georgia

- Co-authored a peer-reviewed paper published at FLAIRS-39 (2026) investigating whether team synergy can be extracted from team composition alone for sport outcome prediction
- Designed a Siamese Transformer model treating players as tokens, capturing intra-team interactions via shared self-attention, and predicting win probabilities from coordinate-wise team representation differences
- Tested across a variety of real and synthetic datasets, achieving 63.3% accuracy on predicting NBA game outcomes and 96.8% on a synthetic league of legends dataset

Data and AI Intern — May 2024 — August 2024
Premise Health — Nashville, Tennessee

- Developed a predictive model for the numbers of visits across hundreds of the company's health centers. Worked with Databrick's NLP capabilities to allow no-code users to access our forecasts using generated SQL queries.
- Documented my 12-week experience with the company's new AI infrastructure to help determine where ML could help the most and what tools are needed to carry out a large-scale ML project.

Computer Science Teaching Assistant — August 2023 — December 2023
Berry College Department of Computer Science — Mount Berry, Georgia

- Provided comprehensive guidance and engaging demonstrations in CSC 120, an introductory computer science course, aiding students through lectures and lab sessions while ensuring their grasp of the course's content
- Demonstrated leadership by conducting lab sessions and delivering lectures during my professor's absence, ensuring the seamless continuation of the curriculum

Projects

Local Search for Minimum Vertex Cover — January 2026 — April 2026
Graduate Algorithms Project

- Designed and implemented a hybrid NuMVC / Simulated-Annealing local search algorithm for the MVC problem, combining edge-weighting, conflict-change tracking, and an adaptive cooling schedule with Metropolis acceptance criterion
- Benchmarked across 20 random seeds on large real-world graphs, matching the best-known cover size on every run of one benchmark in under 0.05 seconds and landed within 0.7% of best-known on a second, harder benchmark

Mechanistic Interpretability Research Project — January 2025 — May 2025
Independent Research

- Used TransformerLens to investigate how GPT-2 forms acronyms using attention and MLP circuits; ran activation patching experiments to isolate components responsible for token prediction behavior
- Designed a corruption-based evaluation framework to analyze how attention heads process capitalized context tokens and their influence on acronym completion

INTERESTS

- Chess, Golf, Skiing, Hiking, Guitar, Classic Novels, Rock Music